11-02 第一篇

Volcanoes: Earth's Fiery Forces

Deep beneath the earth's surface, temperatures are extremely high. Rock melts into magma, and pressure builds up over time. When this pressure becomes too great, magma forces its way to the surface through cracks and vents, creating a volcanic eruption. Volcanoes have shaped the landscape of our planet for billions of years, forming mountains, islands, and even new land. [1] Volcanoes are among the most powerful natural forces on Earth, and understanding how they work helps scientists predict eruptions and protect communities that live nearby.

There are different types of volcanoes. Shield volcanoes, like those in Hawaii, produce gentle flows of lava that spread over wide areas. Stratovolcanoes, such as Mount Fuji in Japan, are steeper and more explosive, sending ash and rock high into the sky. Cinder cone volcanoes are smaller and often form quickly from eruptions of lava fragments. [2] The type of volcano depends on the kind of magma beneath it and how gases escape, which determines whether an eruption is calm or violent.

Volcanic eruptions can be devastating. Lava flows destroy everything in their path, and ash clouds can block sunlight for days. Mudflows, called lahars, rush down the slopes and bury entire villages. The eruption of Mount Vesuvius in 79 AD famously destroyed the Roman cities of Pompeii and Herculaneum. [3] Despite the destruction they cause, volcanic eruptions are a natural part of how our planet releases internal heat and reshapes its surface over time.

Volcanoes also bring benefits. Volcanic soil is rich in minerals, making surrounding areas excellent for farming. Geothermal energy from volcanic regions provides clean electricity in countries like Iceland and New Zealand. Volcanic rock is used in construction, and hot springs attract tourists. [4] The same forces that make volcanoes dangerous also make the land around them fertile and full of resources, which is why people have lived near volcanoes throughout history.

Today, scientists monitor active volcanoes using seismometers, satellite images, and gas sensors. Early warning systems give people time to evacuate before major eruptions. While we cannot stop volcanoes from erupting, we can prepare and reduce the risk to human life. [5] Advances in volcano monitoring technology continue to save lives, turning our growing knowledge of these fiery giants into practical tools for disaster prevention.

选项

A. Volcanoes are among the most powerful natural forces on Earth, and understanding how they work helps scientists predict eruptions and protect communities that live

nearby. B. The type of volcano depends on the kind of magma beneath it and how gases escape, which determines whether an eruption is calm or violent. C. Despite the destruction they cause, volcanic eruptions are a natural part of how our planet releases internal heat and reshapes its surface over time. D. The same forces that make volcanoes dangerous also make the land around them fertile and full of resources, which is why people have lived near volcanoes throughout history. E. Advances in volcano monitoring technology continue to save lives, turning our growing knowledge of these fiery giants into practical tools for disaster prevention. F. For example, the 1980 eruption of Mount St. Helens in the United States reduced the mountain's height by about 400 meters and flattened forests over a huge area.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：空前描述地球深处高温、岩浆形成、压力导致喷发、火山塑造地貌数十亿年，空后需概括火山的重要性及研究意义。A的"most powerful natural forces on Earth"与"shaped the landscape... for billions of years"呼应，"understanding how they work helps scientists predict eruptions"与下文类型分析自然衔接。
- **[2] B**：
  - **线索**：空前介绍三种火山类型——盾状火山（夏威夷）、层状火山（富士山）、渣锥火山，空后需解释类型差异的原因。B的"The type of volcano depends on the kind of magma beneath it and how gases escape"解释了分类依据，"determines whether an eruption is calm or violent"与盾状火山的温和与层状火山的爆炸性形成对应。
- **[3] C**：
  - **线索**：空前说火山喷发的破坏——熔岩摧毁、灰云遮阳、泥流掩埋、庞贝城毁灭，空后需转折说明火山喷发的自然属性。C的"Despite the destruction they cause"承接前文破坏，"a natural part of how our planet releases internal heat and reshapes its surface"提供宏观视角，形成转折。
- **[4] D**：
  - **线索**：空前说火山的好处——土壤肥沃、地热能源、火山岩建材、温泉旅游，空后需总结火山危险与益处的辩证关系。D的"The same forces that make volcanoes dangerous also make the land around them fertile"与前文好处呼应，"why people have lived near volcanoes throughout history"解释了人类选择。

- [5] E：
    - **线索**：空前说科学家用地震仪、卫星、气体传感器监测火山、预警系统提供疏散时间，空后需总结科技监测的价值。E的"Advances in volcano monitoring technology continue to save lives"与前文监测手段呼应，"turning our growing knowledge... into practical tools for disaster prevention"是全文收尾。
- **干扰项分析**：
    - **F. For example, the 1980 eruption of Mount St. Helens in the United States reduced the mountain's height by about 400 meters and flattened forests over a huge area.**：以For example开头，为具体火山喷发案例（圣海伦斯火山）。各空均为概括或过渡句；[3]易被误选因涉及破坏，但该空需转折说明火山的自然属性，而非另一个破坏案例。